Introduction

This document presents the testing and evaluation process developed for the MamaMind project. The purpose of this work was to assess how different approaches could identify patterns of emotional distress in journal-style text related to postpartum wellbeing. Because emotional distress is not always expressed directly, the evaluation focused on comparing methods that could detect both explicit and subtle indicators of concern.

The testing process was designed to compare rule-based logic, sentiment analysis, AI-assisted classification, and hybrid scoring methods in order to understand their performance, reliability, and practical value within the MamaMind system. Structured CSV test data was used to simulate different emotional expressions and distress scenarios, making it possible to review outputs in a more consistent and measurable way.

This document outlines the evaluation approach, the tools used, the data structure, and the main findings that informed the development of the MamaMind user-facing support experience.

TESTING & EVALUATION

This section explains how MamaMind was tested to check both technical functionality and risk assessment performance. It covers manual feature testing, AWS integration testing, scenario-based testing, dataset-based evaluation, and analysis of the final results.

Testing Approach

The testing approach combined manual, integration, and dataset-based evaluation to assess both functionality and system performance. Manual testing was used to verify core features, including secure sign-in, journal submission, retrieval of entries, sentiment and risk feedback, EPDS-style check-ins, explainable outputs, support guidance, and the Petra chatbot. Different journal scenarios (low to urgent risk) were tested to confirm appropriate system responses.

Integration testing ensured correct communication between the React frontend and AWS services, including authentication with Amazon Cognito, API requests via Amazon API Gateway, processing with Lambda, storage in Amazon DynamoDB, analysis using Amazon Comprehend and Amazon Bedrock, and chatbot interaction through Amazon Lex. Amazon CloudWatch was used for debugging.

Dataset-based evaluation used a CSV dataset of 200 journal entries stored in S3. Lambda functions processed each entry, compared predictions with expected labels, and calculated accuracy, macro precision, and recall, enabling consistent and repeatable evaluation of risk logic.

testset-final-website-epds

text	expected sentiment	expected text risk score	epds score	expected final risk score
I felt close to my baby today and that made the tiredness easier to handle.	POSITIVE	Low Risk	3	Low Risk
I am adjusting slowly but I feel safe and supported.	POSITIVE	Low Risk	4	Low Risk
I am tired of feeling like I should be happier than I am.	NEGATIVE	Medium Risk	8	Medium Risk
I feel blank and distant sometimes and that scares me.	NEGATIVE	Medium Risk	9	Medium Risk
I feel trapped in my own head and I do not know how to keep doing this every day.	NEGATIVE	High Risk	9	High Risk
I love my baby but I feel so detached that it scares me and makes me feel like a terrible mother.	NEGATIVE	High Risk	9	High Risk
I cannot sleep even when the baby sleeps because my thoughts will not stop.	NEGATIVE	High Risk	9	High Risk
I feel unsafe and I need help now.	NEGATIVE	High Risk	18	High Risk
My entry sounds calm but my check-in score shows I have been deeply unhappy and unable to cope.	NEGATIVE	Low Risk	13	High Risk
I said today was fine but my EPDS answers show I have been crying often and not enjoying anything.	NEGATIVE	Low Risk	14	High Risk
I wrote a short neutral entry but my check-in score suggests significant distress.	NEUTRAL	Low Risk	15	High Risk
I am still functioning but the questionnaire shows I have been overwhelmed nearly every day.	NEGATIVE	Medium Risk	13	High Risk
I feel anxious and guilty, and my EPDS score suggests I should speak to someone soon.	NEGATIVE	Medium Risk	14	High Risk
I do not feel immediately unsafe, but my check-in score is high and I feel unable to cope.	NEGATIVE	Medium Risk	15	High Risk
The journal entry is not dramatic, but my answers show I am struggling with sadness, sleep, and crying.	NEGATIVE	Medium Risk	13	High Risk
I am pretending to cope but the EPDS questions made me realise how heavy everything has become.	NEGATIVE	Medium Risk	14	High Risk
I feel disconnected and my check-in score shows this has been happening often.	NEGATIVE	Medium Risk	13	High Risk
I keep saying I am okay, but my EPDS score suggests I need professional support.	NEGATIVE	Medium Risk	15	High Risk

Figure 1 DataSet snip

Comparative Logic Evaluation

Comparative logic evaluation was used to compare four different approaches to risk assessment. This was done to understand whether the system performed better with basic sentiment analysis, contextual AI classification, or the final combined website logic.

The same 200-row dataset was used for all four logic versions to make the comparison fair. Each Lambda evaluation function processed the same journal entries and compared the predicted results against the expected labels.

The four logic versions were:

Logic 1	Comprehend + Clear Rules	Uses Amazon Comprehend to detect sentiment, then applies rule-based checks to classify risk.
Logic 2	Bedrock Risk + Clear Rules	Uses clear safety rules first, then Amazon Bedrock to classify contextual postpartum risk.
Logic 3	Bedrock Sentiment + Bedrock Risk + Clear Rules	Uses Amazon Bedrock to classify both sentiment and contextual risk from the journal text.
Logic 4	Final Website Logic	Combines Comprehend sentiment, Bedrock risk classification, urgent safety rules, and EPDS score to produce the final risk level.

Figure 2 Testing Logic Table

Logic 1 acted as the baseline approach. Logic 2 tested whether Bedrock could improve contextual risk classification. Logic 3 tested whether Bedrock could perform both sentiment and risk analysis. Logic 4 tested the actual risk logic used in the final MamaMind website.

This comparison helped identify which approach produced the strongest risk assessment performance.

Evaluation Metrics

The evaluation used accuracy, macro precision, and macro recall to measure how well each risk logic performed. These metrics were calculated by comparing the predicted outputs from the Lambda evaluation functions against the expected labels in the dataset.

Results

The results showed that the risk assessment performance improved as the logic became more contextual and included structured screening.

Metric	Logic1	Logic2	Logic3	Website Logic
Sentiment Accuracy	78.50%	n/a	79.00%	78.50%
Risk Accuracy	75.00%	89.50%	91.00%	96.00%
Sentiment Precision	47.20%	n/a	50%	47.20%
Sentiment Recall	39.20%	n/a	39.50%	39.25%
Risk Precision	71.10%	91.57%	92.50%	96.49%
Risk Recall	74.01%	81.58%	84.75%	95.70%

Figure 3 Results table.

For comparative analysis, the mapped risk results were used for the Bedrock-based logic versions. This is because Bedrock could return `Urgent` as a separate category, while the dataset used `High Risk` as the highest risk label. Since the application treats `Urgent` as a high-risk case with an additional safety flag, `Urgent` was mapped to `High Risk` to allow a fair comparison across all logic versions. The full Lambda test outputs for each logic version are provided in **Appendix 1**.

Interpretation

The results show a clear improvement in risk assessment performance as the logic became more contextual and included structured screening. Logic 1 achieved **75.00% risk accuracy**, which shows that Comprehend sentiment analysis and clear rules provided a useful baseline. However, this result also suggests that sentiment analysis alone is limited for postpartum wellbeing because risk is not always expressed through obviously negative language.

Logic 1 produced the lowest risk accuracy, showing that Comprehend sentiment and rules provided a useful baseline but were limited for more subtle postpartum risk language. Logic 2 improved

performance by using Bedrock for contextual risk classification. Logic 3 improved slightly further by asking Bedrock to analyse both sentiment and risk. Logic 4 achieved the strongest result because it represented the final website logic, combining Comprehend, Bedrock, urgent safety rules, and EPDS score.

Logic 2 improved to **89.50% mapped risk accuracy** by using Bedrock for contextual risk classification. This indicates that Bedrock was better at recognising subtle or indirect postpartum risk signals, such as masking feelings, emotional numbness, avoidance, guilt, or mixed emotions.

Logic 3 achieved **91.00% mapped risk accuracy**, which suggests that Bedrock performed strongly when asked to analyse both sentiment and contextual risk. This shows that a large language model can be useful for interpreting complex journal text beyond simple sentiment labels.

Logic 4 achieved the strongest result, with **96.00% final risk accuracy**, **96.49% final risk macro precision**, and 95.70% final risk macro recall. This was the most important result because Logic 4 represented the actual website logic, combining Comprehend sentiment, Bedrock contextual risk classification, EPDS score, and urgent safety rules.

The improvement from **88.50% text risk accuracy** to **96.00% final risk accuracy** in Logic 4 shows that the EPDS-style check-in strengthened the final risk assessment. Overall, the results support the final design decision to combine AI-based contextual analysis, structured screening, and clear safeguarding rules rather than relying on sentiment analysis alone.

Limitations

Although the evaluation showed strong results, several limitations exist. The dataset was synthetic and manually labelled, so it may not reflect real postpartum journal entries, which can be more complex and context-dependent. EPDS scores were also generated for testing rather than collected from real users, meaning the system is not clinically validated despite strong performance.

Manual labelling introduces subjectivity, as interpretations of emotional distress may vary. The system also relies on AWS AI services such as Amazon Bedrock, where outputs can vary, requiring safety guardrails and fixed rules for urgent cases.

Finally, the evaluation used only 200 examples. Larger datasets, clinical validation, real user testing and further safety assessment are needed before real-world deployment.

CONCLUSION

MamaMind successfully demonstrates a cloud-based postpartum wellbeing journaling prototype that combines React with AWS services. The system allows users to sign in securely, write journal entries, receive sentiment and contextual risk feedback, complete an EPDS-style check-in, view explainable recommendations, and optionally interact with Petra chatbot.

The key finding from the project was that sentiment analysis alone was not enough for reliable postpartum risk assessment. The evaluation showed that the strongest performance came from the final website logic, which combined Comprehend sentiment, Bedrock contextual risk classification, EPDS score, and clear safety rules. Logic 4 achieved **96.00% final risk accuracy**, showing that structured screening and contextual AI improved the assessment compared with simpler approaches.

From a development perspective, the project was successful because it evolved from a simple journal app into a full cloud-based prototype with authentication, storage, AI analysis, chatbot support, deployment, and evaluation. A major challenge was integrating several AWS services and resolving issues around Lambda permissions, Lex configuration, Bedrock access, and frontend-backend communication. These challenges improved my understanding of cloud architecture and debugging in a serverless environment.

The project also highlighted the importance of responsible computing practice. Since the application deals with sensitive wellbeing data, it was important to avoid diagnosis, include explainability, provide crisis guidance, and apply safety guardrails around AI-generated outputs. This helped me understand that technical functionality is not enough in health-related systems; ethical design, security, usability, and safeguarding must also be considered.

Overall, MamaMind achieved its aim as an academic prototype, but it would require further work before real-world use. Future improvements would include testing with clinically reviewed datasets, involving healthcare professionals, strengthening privacy controls, adding multilingual support, improving long-term trend analysis, and conducting user testing with mothers.